

To bridge this knowledge gap, this research project investigates the flow behaviour and performance of tube banks at extended transverse spacings using 2D Unsteady Reynolds-averaged Navier-Stokes (URANS) computational fluid dynamics (CFD) simulations.

Background

- **Hydrogen Aviation:** Liquid hydrogen produces three times more energy per kilogram than kerosene while producing no CO₂ emissions at the point of use.
- **Extended Spacing Challenge:** Low pressure drop requirements in the Lockheed hydrogen aircraft studies of the 1970s and 1980s used wide tube spacings ($X_t^* = 6$), outside the range of validity of existing correlations.
- **Literature Limitation:** Standard correlations (Kays & London $X_t^* = 1.5$, Gaddis & Gnielinski $X_t^* \leq 3$) overpredict performance when extrapolated.
- **Numerical Strategy:** 2D URANS balances computational cost and resolves transient vortex shedding across multi-lane domains.

Simulation Setup

CFD simulations were performed with unsteady RANS using $k-\omega$ SST turbulence model on Star-CCM+. Convergence was tracked with 5 residuals and by tracking key parameters of interest. The computational domain for the wide spaced tube bank geometry studied is shown in Fig.1 and detailed in Table.1, and the boundary conditions used are detailed in the table and in Fig.2. The boundary conditions and geometry of the domain are matched to the experiments by Kays and London. An unstructured mesh was generated for each geometry, and prism layers were chosen to achieve $y^+ < 1$ on wetted surfaces.

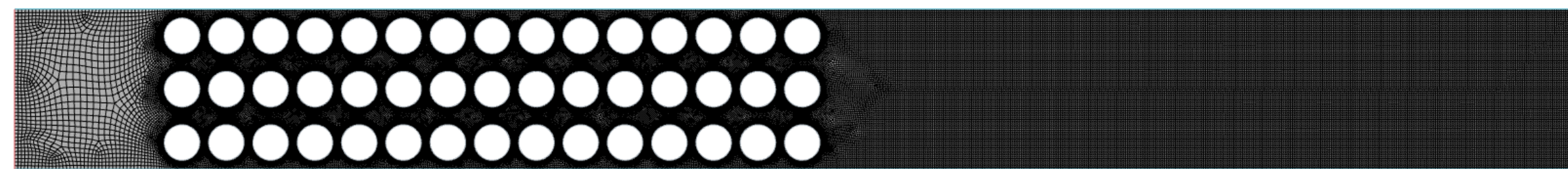


Figure 1. $d_o/5$ base cell size mesh.

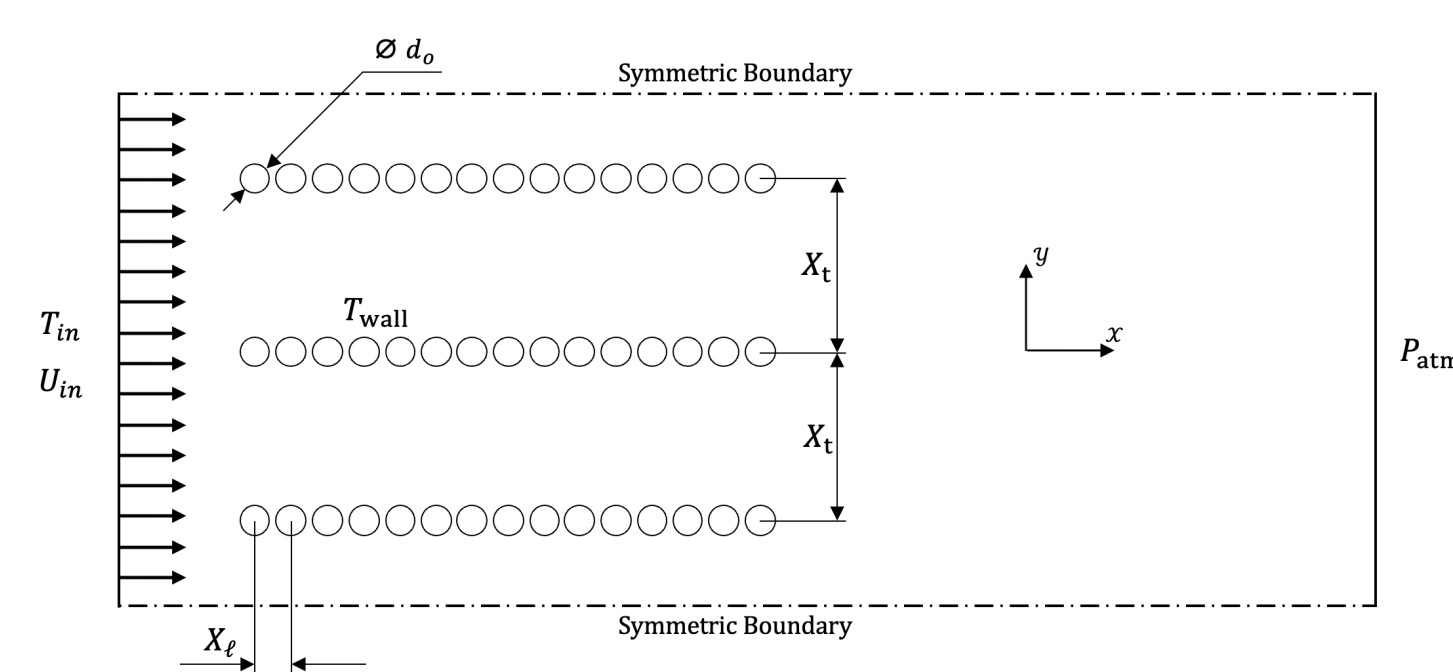


Figure 2. Diagram of the simulation domain.

Parameter	Description	Dimension
d_o	Tube Outer Diameter	0.00635 m
X_t	Transverse Spacing	$6 d_o$
X_ℓ	Longitudinal Spacing	$1.25 d_o$
T_{in}	Inlet Temperature	300 K
T_{wall}	Tube Wall Temperature	373.15 K
P_{atm}	Atmospheric Pressure	101,325 Pa
U_{in}	Inlet Velocity	Variable

Table 1. Domain parameters.

Mesh Independence Study

A mesh independence study was performed to calculate the minimum mesh size required, in order to reduce the run time of simulations, while still providing results that minimise the errors introduced from the mesh. The base cell size was decreased from $d_o/2$ to $d_o/10$, with the flow parameters were evaluated and compared to the finest mesh produced, Fig.3. It was desirable for the mesh to have a percentage difference of less than 1% when compared to the finest mesh, as such the base size of $d_o/5$ was chosen, shown in Fig.1. This mesh size increased the run time of the simulations for the $d_o/5$ mesh to 6 hours compared to 3 hours for the $d_o/2$ mesh.

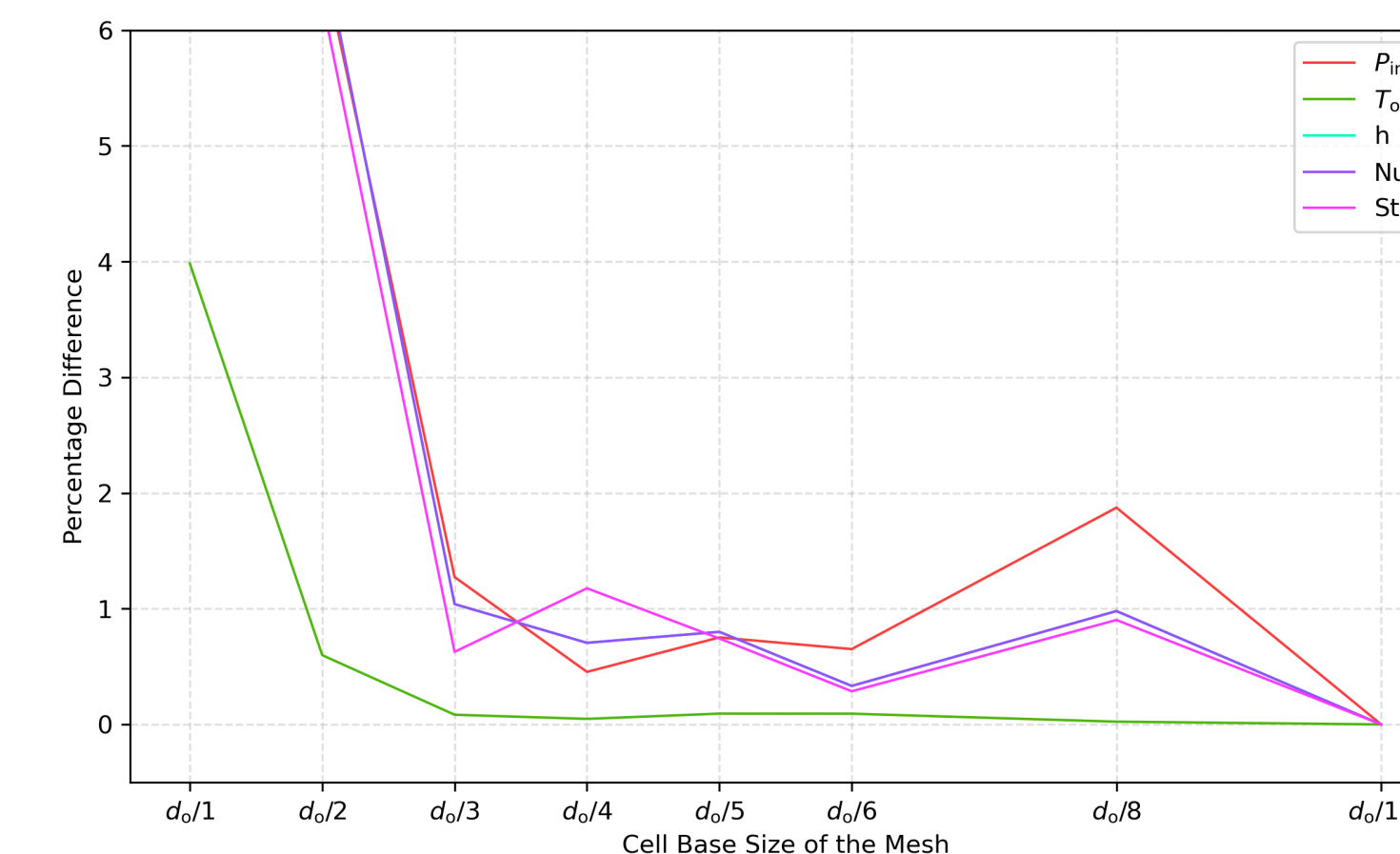


Figure 3. Key parameters compared to finest mesh

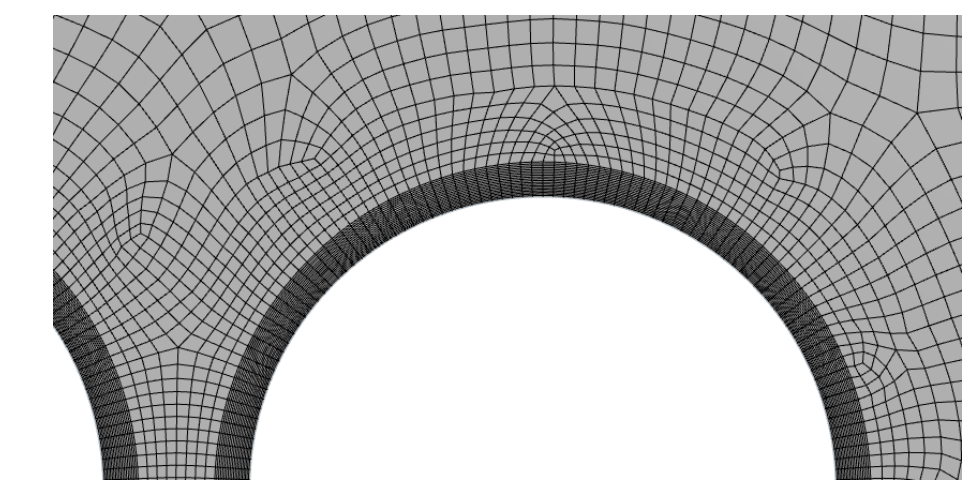


Figure 4. Prism layers of $d_o/5$ mesh around the tube walls.

Results and Discussion

The simulation results for the extended transverse spacing of $X_t = 6d_o$ are plotted below in Fig.5 showing the dimensionless heat transfer and Fig.6 showing the dimensionless viscous shear.

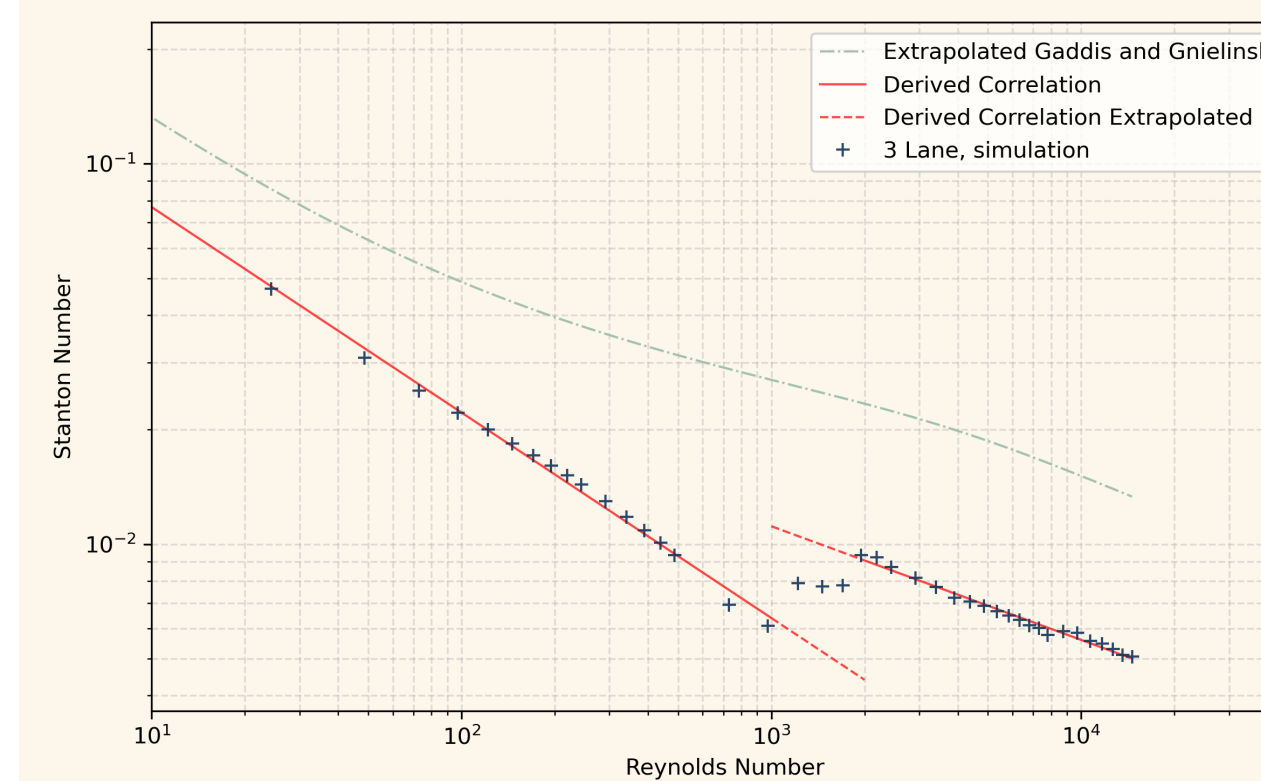


Figure 5. Dimensionless heat transfer

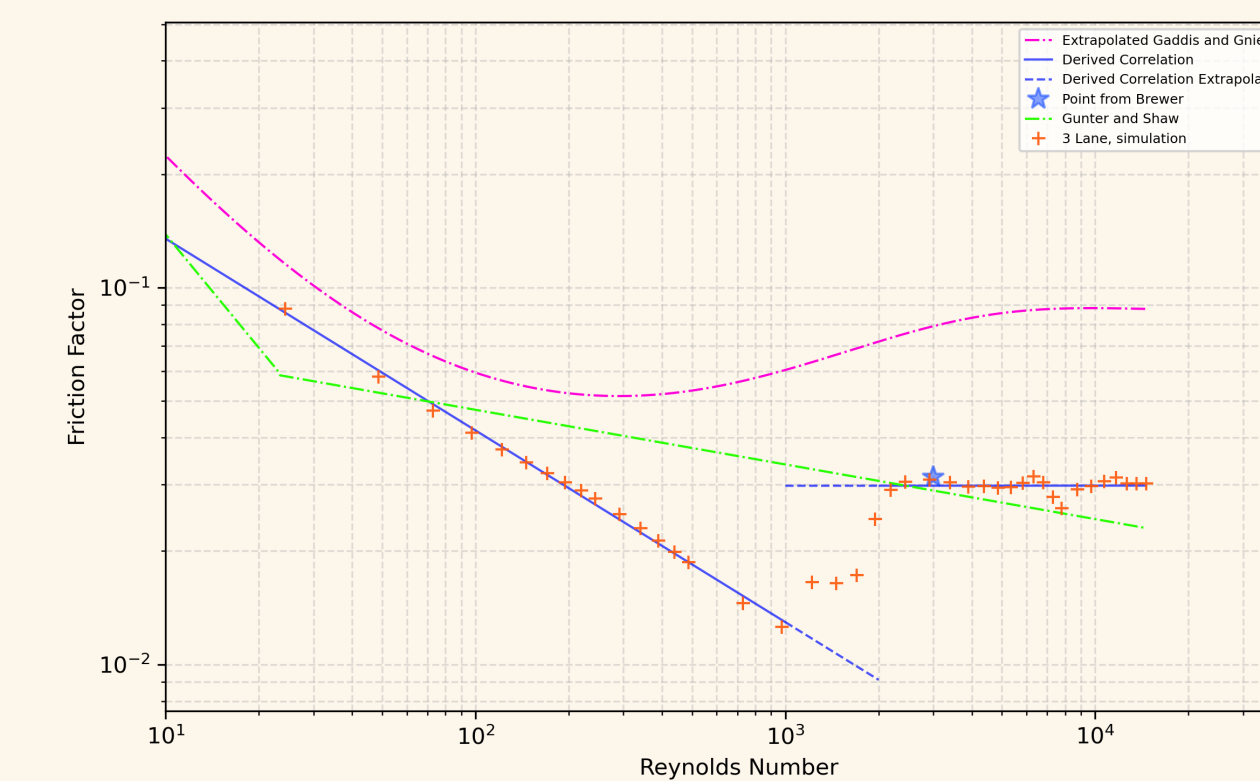


Figure 6. Dimensionless viscous shear (friction)

It is clear that the extrapolated Gaddis and Gnielinski correlation is not good at predicting the heat transfer of friction for the wide spacing, which was to be expected as these should only be used for transverse spacings up to $X_t^* = 3$. However, the generic Gunter and Shaw correlation for the friction is incidentally quite accurate at the Reynolds number of the Brewer design point.

To compliment the literature, it was desirable to produce a set of correlations that could be used for calculating the Stanton number and friction factor across different Reynolds numbers. After computing the derived correlations, they were plotted alongside the simulation data, with the Stanton number correlation plotted in Fig.5, and the friction factor correlation plotted in Fig.6.

$$\text{St}(\text{Re}_d, \text{Pr}) = \begin{cases} 0.298 \text{Re}_d^{-0.540} \text{Pr}^{1/3}, & 20 \leq \text{Re}_d \leq 1,000 \\ 0.0980 \text{Re}_d^{-0.299} \text{Pr}^{1/3}, & 2,000 \leq \text{Re}_d \leq 10,000 \end{cases} \quad (1)$$

$$f(\text{Re}_d) = \begin{cases} 0.435 \text{Re}_d^{-0.509}, & 20 \leq \text{Re}_d \leq 1,000 \\ 0.0297, & 2,000 < \text{Re}_d < 10,000 \end{cases} \quad (2)$$

To better understand the unsteady flow behaviour for incline tube banks, a 3 lane simulation was ran at different Reynolds numbers, these are shown in Fig.7. In the progression of Reynolds number, you can see the change from structured vortex streets visible at $Re_d = 150$ to the transitional at $Re_d = 1,460$, to the final high Reynolds case at $Re_d = 14,600$ where the flow around the tube bank has oscillatory structure forming before the wake and within the tube bank. This analysis was useful to understand the difficulties of using CFD for tube bank heat exchanger predictions.

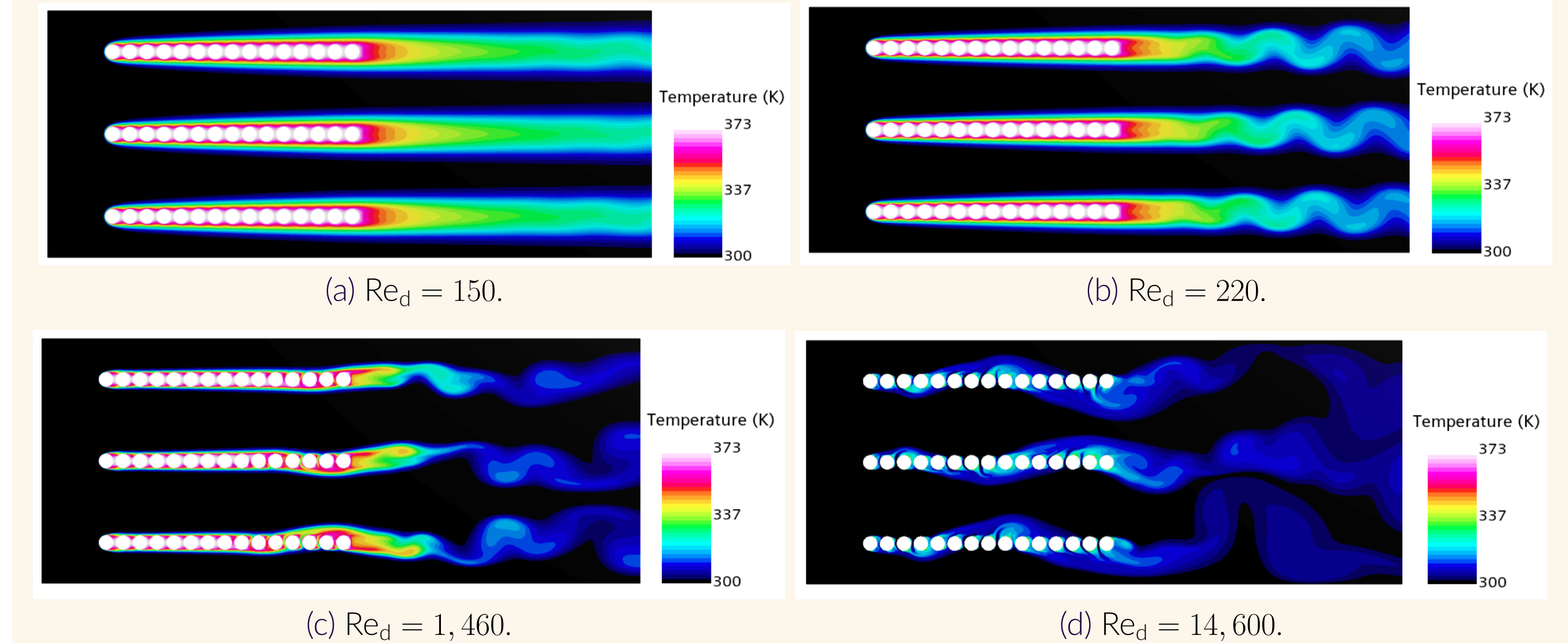


Figure 7. Inline $X_t^* = 6$, $X_\ell^* = 1.25$ temperature scenes for different Reynolds number.

Conclusion

This research project evaluated heat exchanger performance at extended transverse spacings ($X_t^* = 6$) for recuperated hydrogen turbofan aircraft:

- Turbulent URANS modelling with at least three tube lanes was shown to capture the increase in friction factor and Stanton number seen in low transverse spacing published experiments: vortex streets and coalescing jets are the underlying physical structure and instability.
- The three-lane simulations successfully replicated the single data point from Brewer's hydrogen aircraft research, which was also well predicted by Gunter and Shaw's correlation for both transverse and inline tube banks.
- That correlation did not however account for inline tube bank's characteristic transition region of low friction factor and Stanton number which is shown in experimental data by Kays and London. The region was found to have a lower friction factor by about 0.01 to 0.02, corresponding to a 25% to 50% lower value than that predicted by the correlation.
- New piecewise log-linear correlations, Eqn.1 and Eqn.2, were successfully derived from the simulation data. These correlations offer predictive models tailored specifically for inline configurations with a transverse spacing of $X_t^* = 6$.

References

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